ECE 584 Class Project Final Report Template

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Abstract

Balancing an inverted single and double pendulum system on a cart is a classical problem in the field of controls and cyber physical systems. Single and double pendulums are examples of nonlinear systems, and the double pendulum specifically is a chaotic system which is more difficult to control. While the control of these systems has been thoroughly studied, the verification of such cyberphysical nonlinear systems is an ongoing field of research, with single and double inverted pendulum controllers being common benchmarks in this area. In this paper, we examine Linear Quadratic Regulator controllers for single and double inverted pendulum systems and implement these controllers in the reachability analysis tool Flow* (3). We then formulate strategies to verify the stability of these controllers using Flow*. We present a straightforward method of completely verifying the single inverted pendulum controller for an initial set, as well as a method for verifying the double inverted pendulum controller. We present our verification results for both the single and double inverted pendulum controllers, and we end with a discussion on future directions.

1. Introduction (1 page)

The study of nonlinear cyberphysical systems is important due to the place they have in the real world. Cyberphysical systems become increasingly relevant in our lives, from autopilot controls to self-driving cars. The control of cyberphysical systems is an ongoing field of study in its own right, but the verification of such controllers is an area that still has a lot of work to be done. A couple of classic controls examples are the single inverted pendulum and double inverted pendulum cart controllers, which we aim to implement and verify using the tool Flow*. In doing so, we will build a greater understanding of strategies on how to verify nonlinear cyberphysical systems.

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Background

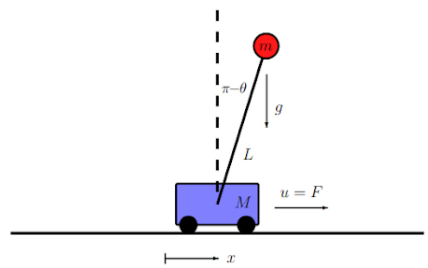

Figure 1. Diagram of SIPC System

Single Inverted Pendulum Controller The Single Inverted Pendulum Controller (SIPC) system we will be using is depicted above (2). The state space consists of four variables: x, v, θ, ω, where x and v are the position and velocity of the cart, θ is the angle of the pendulum arm from the vertical position, and $omega$ is the angular velocity of the pendulum arm. There are also a number of variables we define, such as $M = 1.0$ for cart mass, $m = 0.1$ for mass of the end of the pendulum, $L = 0.5$ for length of the pendulum arm, $g = 9.81$ for the force of gravity, and $\delta = 1$ for the damping coefficient. The state space is governed by the following dynamic equations (2):

$$
\begin{aligned}
\dot{x} &= v \\
\dot{v} &= \frac{-m^2L^2g\cos(\theta)\sin(\theta) + mL^2\omega^2(mL\omega^2\sin(\theta) - \delta v) + mL^2u}{mL^2(M + m(1-\cos(\theta)^2))} \\
\dot{\theta} &= \omega \\
\dot{\omega} &= \frac{(m+M)mgL\sin(\theta) - mL\cos(\theta)(mL\omega^2\sin(\theta) - \delta v) + mL\cos(\theta)u}{mL^2(M + m(1-\cos(\theta)^2))}
\end{aligned} \tag{1}
$$

The system also includes an input variable u, which will be an input force applied horizontally to the cart. This will be driven by a controller, which will be touched upon later.

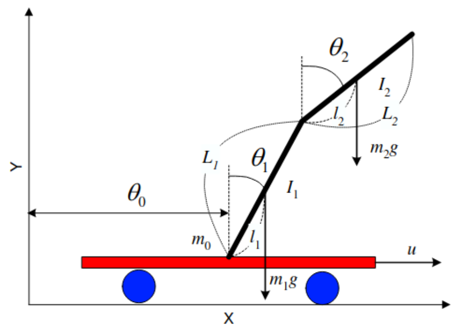

Figure 2. Diagram of DIPC System

Double Inverted Pendulum Controller The Double Inverted Pendulum Controller (DIPC) system we'll be using is shown above (1, 4). This system has a state space that consists of six variables, shown here as $(\theta_0, \theta_1, \theta_2, \dot{\theta_0}, \dot{\theta_1}, \dot{\theta_2})$. θ_0 and $\dot{\theta_0}$ are the position and velocity of the cart, θ_1 and $\dot{\theta_1}$ are the angle from vertical and angular velocity of the lower arm of the double pendulum, and θ_2 and $\dot{\theta_2}$ are the angle from vertical and angular velocity of the upper arm of the double pendulum. User-defined variables include $m_0 = 1.5$ for the mass of the cart, $m_1 = m_2 = 0.5$ for the masses of the lower and upper arm, $L_1 = L_2 = 0.5$ for the lengths of the lower and upper arms, and $g = 9.8$ for gravitational acceleration. The image also shows variables l_1 and l_2 for the length from base of the center of gravity, and I_1 and I_2 for the moments of inertia for the two arms. To simplify the system though, it is assumed that the center of gravity is at the midpoint of each arm, so these variables don't need to be defined.

The equations of motion (1) for this system can be written in a matrix form:

$$\mathbf{D}(\theta)\ddot{\theta} + \mathbf{C}(\theta, \dot{\theta})\dot{\theta} + \mathbf{G}(\theta) = \mathbf{H}u$$

where

$$\mathbf{D}(\theta) = \begin{pmatrix} d_1 & d_2\cos(\theta_1) & d_3\cos(\theta_2) \\ d_2\cos(\theta_1) & d_4 & d_5\cos(\theta_1 - \theta_2) \\ d_3\cos(\theta_2) & d_5\cos(\theta_1 - \theta_2) & d_6 \end{pmatrix}$$
$$\mathbf{C}(\theta, \dot{\theta}) = \begin{pmatrix} 0 & -d_2\sin(\theta_1)\dot{\theta_1} & -d_3\sin(\theta_2)\dot{\theta_2} \\ 0 & 0 & d_5\sin(\theta_1 - \theta_2)\dot{\theta_2} \\ 0 & -d_5\sin(\theta_1 - \theta_2)\dot{\theta_1} & 0 \end{pmatrix}$$
$$\mathbf{G}(\theta) = \begin{pmatrix} 0 \\ -f_1\sin(\theta_1) \\ -f_2\sin(\theta_2) \end{pmatrix}$$
$$\mathbf{H} = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}^T \tag{2}$$

The coefficients in these matrices are defined as:

$$\begin{aligned} d_1 &= m_0 + m_1 + m_2 \\ d_2 &= (\frac{1}{2}m_1 + m_2)L_1 \\ d_3 &= \frac{1}{2}m_2 L_2 \\ d_4 &= (\frac{1}{3}m_1 + m_2)L_1^2 \\ d_5 &= \frac{1}{2}m_2 L_1 L_2 \\ d_6 &= \frac{1}{3}m_2 L_2^2 \\ f_1 &= (\frac{1}{2}m_1 + m_2)L_1 g \\ f_2 &= \frac{1}{2}m_2 L_2 g \end{aligned}$$

From this, we can derive the state space dynamic equations:

$$\dot{\mathbf{x}} = \begin{pmatrix} \mathbf{0} & \mathbf{I} \\ \mathbf{0} & -\mathbf{D}^{-1}\mathbf{C} \end{pmatrix}\mathbf{x} + \begin{pmatrix} \mathbf{0} \\ -\mathbf{D}^{-1}\mathbf{G} \end{pmatrix} + \begin{pmatrix} \mathbf{0} \\ \mathbf{D}^{-1}\mathbf{H} \end{pmatrix}u \tag{3}$$

Again, this system also includes an input variable [u], which is the force applied horizontally to the cart by the controller.

Linear Quadratic Regulator The remaining question is how to design an optimal controller for these systems. One technique is centered around minimizing a cost function:

$$J_t = \sum_{k=t}^{t_{final}} L_k(\mathbf{x}_k, \mathbf{u}_k)$$

This represents the total accumulated cost for every state $\mathbf{x}_k$ and control $\mathbf{u}_k$ from time t to t_{final}. L_k in this case corresponds to Linear Quadratic cost:

$$L_k(\mathbf{x}_k, \mathbf{u}_k) = \mathbf{x}_k^T\mathbf{Q}\mathbf{x}_k + \mathbf{u}_k^T\mathbf{R}\mathbf{u}_k \tag{4}$$

Here, the **Q** matrix represents the cost of deviation from the desired state, and the **R** matrix represents the cost of controller input. Both are user-defined. The dynamics equations for SIPC and DIPC can then be linearized around **x=0** as such:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u \tag{5}$$

To implement digital control, matrices A and B are approximately discretized into $\mathbf{\Phi} \approx e^{\mathbf{A}\Delta t}$ and $\mathbf{\Gamma} \approx \mathbf{B}\Delta t$. Optimal digital LQR control is then given by:

$$u_k = -\mathbf{R}^{-1}\mathbf{\Gamma}^T\mathbf{P}\mathbf{x}_k \equiv -\mathbf{K}\mathbf{x}_k$$

The **P** matrix is found by solving the discrete-time algebraic Riccati equation:

$$\mathbf{\Phi}^T[\mathbf{P} - \mathbf{P}\mathbf{\Gamma}(\mathbf{R} + \mathbf{\Gamma}^T\mathbf{P}\mathbf{\Gamma})^{-1}\mathbf{\Gamma}^T\mathbf{P}]\mathbf{\Phi} - \mathbf{P} + \mathbf{Q} = \mathbf{0}$$

The problem. Clearly state the problem you are trying to solve. Important concepts in the problem statement should be defined before they are used. For example, if Graph Neural Network (GNN) or Scan lines are used in the problem statement, then those should be defined in the background paragraph. You are encouraged to use mathematical notations to define the problem precisely and succinctly as long as the notations are defined.

State of the art. What are the existing solutions to this problem? What are their limitations? Why is there a need for a new solution? A more detailed literature review will go in the related work section.

Your approach. What is your proposed solution to the problem? How is it different from the existing solutions?

Contributions Summary of key contributions in this work.

2. Related work (1/2 page)

Discuss the existing work in this area. What are the key contributions of previous studies? How does your work relate to or differ from these prior efforts? You can cite papers and books using (2, 3, 4, 1) and include the references in the final.bib file.

3. Methodology (2-3 pages)

SIPC Implementation Implementing the SIPC system in Flow* is fairly straightforward. Flow* provides the functions and classes necessary to both incorporate the ODEs and the control law of the system and build a testing routine in C++. The first step is to create a Flow* Variables object and to define the necessary variables for the system. This includes both the state space variables and the input variables. For SPIC, we have four state space variables and one input variable, assigned as such:

```
// Declare variables
Variables vars;
int x_id = vars.declareVar("x");
int v_id = vars.declareVar("v");
int theta_id = vars.declareVar("theta");
int omega_id = vars.declareVar("omega");
int u_id = vars.declareVar("u");
```

Figure 3. Declaring variables for SIPC

4. Results (2-3 pages)

Benchmarks and models. What benchmarks and models are used to evaluate your method?

Evaluation metrics. What metrics are used to evaluate the performance of your method?

Results tables/graphs. Present the results of your experiments using tables and graphs. You are encouraged to organize the tables according to the main lessons you want to convey.

Code. Provide links to your code or github repositories. Save your code and scripts in an organized fashion so that the results presented here can be replicated.

References

[1] Alexander Bogdanov. Optimal control of a double inverted pendulum on a cart. Technical Report CSE-04-006, Oregon Health Science University, Portland, Oregon, December 2004.

[2] Steven L. Brunton and J. Nathan Kutz. *Linear Control Theory*, pages 352–353. Brunton Kutz, 2017.

[3] Xin Chen, Erika Abraham, and Sriram Sankaranarayanan. Flow*: An analyzer for non-linear hybrid systems. In Natasha Sharygina and Helmut Veith, editors, *Computer Aided Verification*, Lecture Notes in Computer Science, pages 258–263, Berlin, Heidelberg, 2013. Springer Berlin Heidelberg.

[4] Ian Crowe-Wright. Control theory: Double pendulum inverted on a cart. Master's thesis, The University of New Mexico, Albuquerque, New Mexico, December 2018.